

Quantitative stability and local error exponents for power Hadamard matrices

8 October 2026

Abstract

Let H be a unit-modulus matrix of order $2m$, and measure the largest unnormalized operator-norm orthogonality residual among its entrywise powers $1, \dots, m-1$. For every $m \geq 2$, a residual at most $2^{-24}m^{-3}$ gives an exact solution at dephased maximum-entry distance at most $512\sqrt{\varepsilon/m}$, together with a certified cyclic generalized Hadamard seed and, when needed, a compatible phase rectangle. For odd m , nearest-root rounding itself produces the seed. A rectangle graph computes the entire simultaneous first-order kernel and supplies polynomial local conditioning. We classify second-order liftable velocities and the best local error exponent at every labelled exact point. For the classical quadratic mutually unbiased basis family at odd primes, the graph has diameter at most seven, giving the explicit linear rate $56H_{p-1}\varepsilon/p$. The global square-root exponent is necessary already at order four. General connectedness at odd half-orders and new seed existence remain open. The exact classification and phase-circle geometry used here retain their external attribution; no historical-priority claim is made.

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1 Introduction and notation

The entrywise powers of a complex Hadamard matrix impose a family of coupled orthogonality conditions. Exact solutions may be discrete or may lie on phase circles, so residual-to-distance estimates depend both on arithmetic root structure and on local geometry. This paper develops explicit quantitative estimates under the full power range $1, \dots, m-1$ for matrices of order $2m$. All input entries are required to have *exactly* unit modulus.

Fix an integer $m \geq 2$ and put $n = 2m$. Write M^* for the conjugate transpose and $M^{\circ k}$ for the entrywise k th power, never the ordinary matrix power. Put

$$\mu_m = \{z \in \mathbb{C} : z^m = 1\}, \quad \zeta_m = e^{2\pi i/m}, \quad h = H_{m-1} = \sum_{k=1}^{m-1} \frac{1}{k}.$$

The norms $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_{\text{max}}$ denote respectively the Euclidean operator norm and the maximum absolute entry. For a unit-modulus H , define

$$\varepsilon(H) = \max_{1 \leq k < m} \|H^{\circ k} (H^{\circ k})^* - nI\|_{\text{op}}, \quad K_{ij} = \frac{H_{ij} H_{00}}{H_{i0} H_{0j}}. \quad (1)$$

We retain all row and column labels. The initial row and column of the dephasing K are all ones. Dephasing multiplies every entrywise power by diagonal unitaries, so $\varepsilon(K) = \varepsilon(H)$. For every square M , the eigenvalue multisets of MM^* and M^*M coincide, including zero multiplicities. Thus

$$\|M^*M - nI\|_{\text{op}} = \|MM^* - nI\|_{\text{op}}. \quad (2)$$

In particular every off-diagonal row or column correlation of each required power has modulus at most ε .

Definition 1.1. A cyclic $\text{GH}(m, 2)$ *seed* is a $2m \times 2m$ matrix G with entries in μ_m such that each ratio of distinct rows contains each element of μ_m exactly twice. Unless stated otherwise, seeds are labelled and dephased. The set $\mathcal{X}_m$ consists of all labelled dephased unit-modulus matrices satisfying $\varepsilon = 0$.

A rectangle $Q = R \times S$, with $|R| = |S| = m$ and both initial indices excluded, is *compatible* with G if for every $a \in R$ and $b \notin R$, the multiset

$$\{G_{aj} \overline{G_{bj}} : j \in S\}$$

contains every m th root once. Its complement then has the same property. For $|\alpha| = 1$, write $\Phi_{G,Q}(\alpha) = \alpha G$ on Q and G off Q .

Finite Fourier inversion shows that the seed condition is equivalent to all the required power orthogonalities for a root matrix. Compatibility makes $\Phi_{G,Q}(\alpha)$ exact for every α . The external exact classification states that all exact dephased matrices are seeds or such circle points; odd m admits no compatible rectangle. Appendix A gives the attributed proof and the exact local geometry used below.

Main results. Theorems 3.1 and 3.2 prove the explicit threshold and rounding statement at odd and arbitrary half-order. Proposition 3.3 gives a linear bound away from root intersections, while Proposition 3.4 proves genuine square-root sharpness at $m = 2$. The integer tangent test of Section 4 is converted into a graph-component formula in Theorem 5.2. Graph diameter gives polynomial conditioning and, for the quadratic prime-order family, a uniform diameter-seven bound. Theorem 8.1 classifies second-order admissibility, and Theorem 9.2 determines the best local exponent at each supplied exact point.

Attribution and scope. The exact root/single-rectangle classification, the two-polygon/common-support/projection-trace architecture, and the finite-circle geometry are attributed to the pinned `mio-qwq/math` source [1]. That source in turn identifies OpenAI's order-six argument [2] as its starting point. These two attribution layers are retained throughout; neither the exact classification nor the known seeds are claimed as new here. The linearized phase and ordinary-defect framework is classical [3]. The quadratic seeds are from the classical MUB construction [4, 5]; the present theorem concerns their *simultaneous all-power* kernel. Ordinary first-power isolation and zero ordinary defect are distinct assertions. The results here do not establish the ordinary isolation conjecture discussed in [5].

This manuscript consolidates the corrected research batch of 8 October 2026. An independent AI-assisted analytical review and a limited revision recheck accepted that batch's stated mathematical core. These are not formal verification, professional mathematical peer review, or historical-priority certification. The finite computations are supplementary exact certificates and are not inputs to the infinite-family analytical proofs. The source map records the precise revision and all inherited material.

2 Quantitative polygons and the common rectangle

This section is parity-free. It supplies the quantitative scalar, support, and block estimates used in both global theorems.

2.1 A scalar product identity

Lemma 2.1. *Let $v_1, \dots, v_{2m}$ lie on the unit circle, include 1, and satisfy*

$$\left| \sum_j v_j^k \right| \leq \varepsilon \quad (1 \leq k < m), \quad 0 \leq \varepsilon \leq 1.$$

Put $d = \prod_j v_j$ and $P(z) = \prod_j (z - v_j)$. There is a polynomial E whose coefficient absolute-value sum is at most $4(m-1)\varepsilon$ such that

$$P(z) = (z^m - 1)(z^m - d) + E(z). \quad (3)$$

Consequently, writing $w_j = v_j^m$ and $t = |d - 1|$,

$$|(w_j - 1)(w_j - d)| \leq 4(m-1)\varepsilon, \quad (4)$$

$$|w_j - 1| \leq t + 2\sqrt{(m-1)\varepsilon}. \quad (5)$$

Proof. Let e_ℓ denote the elementary symmetric functions, with $e_0 = 1$. Newton's identities give by induction $|e_\ell| \leq \varepsilon$ for $1 \leq \ell < m$, since

$$\ell|e_\ell| \leq \varepsilon[1 + (\ell-1)\varepsilon] \leq \ell\varepsilon.$$

The unit-modulus complement identity is $e_{2m-\ell} = d\overline{e_\ell}$. Hence

$$P(z) = z^{2m} + cz^m + d + R(z),$$

where R has at most $2(m-1)$ nonzero coefficients, each of modulus at most ε . The included entry 1 gives $P(1) = 0$, so $c = -1 - d - R(1)$. Taking $E(z) = R(z) - R(1)z^m$ proves the coefficient bound and (3). Evaluate at v_j to obtain (4). At least one of $|w_j - 1|, |w_j - d|$ is at most $2\sqrt{(m-1)\varepsilon}$, proving (5). $\square$

2.2 Separated polygons with multiplicities

Lemma 2.2. *Under Lemma 2.1, assume $\varepsilon > 0$ and $t = |d - 1| \geq 32\sqrt{m\varepsilon}$. The entries admit a bijective matching with the $2m$ distinct roots of $(z^m - 1)(z^m - d)$, each matching distance being at most*

$$r_0 = 64\varepsilon/t.$$

Exactly m entries satisfy $|v_j^m - 1| \leq h_0$ and exactly m satisfy $|v_j^m - d| \leq h_0$, where $h_0 = 64m\varepsilon/t$. These classes are disjoint, and an entry equal to 1 belongs to the first.

Proof. The hypotheses give $r_0 \leq t/(8m)$ and $mr_0 \leq 1/4$. On a circle of radius r_0 about a root u of either factor, write $z = u(1 + w)$ with $|w| = r_0$. The binomial estimate

$$|(1 + w)^m - 1 - mw| \leq (mr_0)^2 e^{mr_0}/2$$

implies $mr_0/2 \leq |z^m - u^m| \leq 2mr_0$. The other factor has modulus at least $t - 2mr_0 \geq 3t/4$, so the product has modulus at least

$$(3/8)mr_0t = 24m\varepsilon.$$

On the same circle, Lemma 2.1 gives

$$|E(z)| \leq 4(m - 1)\varepsilon(1 + r_0)^{2m} < 8(m - 1)\varepsilon.$$

Rouché's theorem gives one zero of P , counting multiplicity, in each disk. The disks are disjoint: roots of different factors have distance at least t/m by telescoping their m th powers; roots of the same factor have distance at least $2\sin(\pi/m) \geq 4/m$. The $2m$ disks therefore count all roots. Since matching endpoints are unit, $|v_j^m - u^m| \leq m|v_j - u| \leq h_0$. Finally $h_0 \leq t/16$, proving disjointness and the assertion about the entry 1. $\square$

The bijection, rather than mere proximity to two phases, certifies multiplicity and ensures that later restricted half-sums contain each root once.

2.3 Common active supports

For a dephased K with $\varepsilon > 0$, put

$$L = K^{\circ m}, \quad d_a = \prod_j K_{aj}, \quad c_j = \prod_i K_{ij}, \quad \delta = 32\sqrt{m\varepsilon}, \quad s = \max_a |d_a - 1|. \quad (6)$$

Call a row active if $|d_a - 1| \geq \delta$, and a column active if $|c_j - 1| \geq \delta$. An active row has a support of m entries of L near d_a , with the remaining entries near 1; the errors are at most $\delta/16$. Every entry of an inactive row is within $17\delta/16$ of 1, by (5). The column assertions are identical. The initial all-ones row and column satisfy these estimates trivially; the scalar moment lemmas apply only to the other rows and columns, using ratios with the initial ones.

Lemma 2.3. *All active rows have the same support, and all active columns have the same support.*

Proof. If active rows a, b have different supports S, U , choose $j \in S \setminus U$. Write $x = d_a$, $y = d_b$, and $w = L_{aj}/L_{bj}$. Then $|w - x| \leq \delta/8$. The actual ratio $v_\ell = K_{a\ell}/K_{b\ell}$ includes 1 at column zero and has product exactly x/y . Thus

$$|(w - 1)(w - x/y)| \leq 4(m - 1)\varepsilon \leq \delta^2/256.$$

Both factors on the left are at least $7\delta/8$, because $|x - 1|, |y - 1| \geq \delta$ and $|x - x/y| = |y - 1|$. This is impossible. Transposition gives the column argument. $\square$

Retaining the actual entry product is essential; an approximate cardinality-parity argument would not supply this conclusion.

2.4 A large phase forces a balanced rectangle

Suppose $s > 8\delta$. Choose a maximizing row a , put $x = d_a$, and let S be the common active-row support. Every column in S is active: its entry at row a is farther than $17\delta/16$ from 1. These columns have a common support T of size m , containing a .

No column outside S is active. Otherwise its support also contains a , where its distance from 1 would be at least $15\delta/16$; row a instead bounds that distance by $\delta/16$. For $j \in S$, comparison at row a gives $|c_j - x| \leq \delta/8$. Thus for $i \in T, j \in S$, $|L_{ij} - x| \leq 3\delta/16$, so row i is active. Conversely an active row cannot lie outside T , by comparison in a column of S . The active rows are precisely $R = T$, and $|R| = |S| = m$.

Permute labels for the proof only, placing R^c, S^c first, and write $K = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. We have

$$\begin{array}{l|l} R^c \times S^c & |L_{ij} - 1| \leq 17\delta/16 \\ R^c \times S & |L_{ij} - 1| \leq \delta/16 \\ R \times S^c & |L_{ij} - 1| \leq \delta/16 \\ R \times S & |L_{ij} - x| \leq 3\delta/16. \end{array} \quad (7)$$

For $j \notin S$ and $\ell \in S$, the column ratio $v_i = K_{ij}/K_{i\ell}$ has product $d = c_j/c_\ell$ and separation

$$t = |d - 1| \geq s - 9\delta/8 > s/2 \geq \delta.$$

Its top-half entries satisfy $|v_i^m - 1| \leq 9\delta/8$. In the matching lemma their power error is at most $\delta/16$, and they cannot belong to the d -polygon since $t - \delta/16 > 9\delta/8$. All m top entries match bijectively to μ_m . Their sum has modulus at most $128m\varepsilon/s$. It follows that

$$\|A^*B\|_{\text{op}} \leq 128m^2\varepsilon/s. \quad (8)$$

Only the first-power restricted half-sum is needed for this bound.

3 Global stability and sharpness

3.1 Odd half-order and a half-integer trace

Theorem 3.1 (Odd half-order rounding). *Let $m \geq 3$ be odd and H have exactly unit-modulus entries. If*

$$\varepsilon(H) \leq 2^{-24}m^{-3}, \quad (9)$$

nearest- m th-root rounding of its labelled dephasing K gives a dephased cyclic $\text{GH}(m, 2)$ seed G , and

$$\|K - G\|_{\text{max}} \leq 512\sqrt{\varepsilon/m}. \quad (10)$$

Undoing the same dephasing gives an exact H_{exact} at the same maximum-entry distance from H . If no cyclic $\text{GH}(m, 2)$ seed exists, every such H has $\varepsilon(H) > 2^{-24}m^{-3}$.

Proof. First take $\varepsilon > 0$. If $s > 8\delta$, use the blocks from Section 2.4. The top principal block of $KK^* - 2mI$ gives

$$E = AA^* + BB^* - 2mI, \quad \|E\|_{\text{op}} \leq \varepsilon.$$

Set $P = AA^*/(2m)$. It is positive semidefinite and $\text{tr } P = m/2$ exactly, because all m^2 entries of A have modulus one. Moreover

$$P^2 - P = \frac{AA^*E - A(A^*B)B^*}{(2m)^2}.$$

The bounds $\|A\|_{\text{op}}^2, \|B\|_{\text{op}}^2 \leq 2m + \varepsilon$ and (8) give

$$\|P^2 - P\|_{\text{op}} \leq \frac{2m + \varepsilon}{4m^2} \left(\varepsilon + \frac{128m^2\varepsilon}{s} \right). \quad (11)$$

For every $\lambda \geq 0$, $\text{dist}(\lambda, \{0, 1\}) \leq 2|\lambda(\lambda - 1)|$. Rounding the m eigenvalues of P therefore produces an integer at distance at most $2m\|P^2 - P\|_{\text{op}}$ from $m/2$. Oddness implies

$$\|P^2 - P\|_{\text{op}} \geq \frac{1}{4m}.$$

For $\varepsilon \leq 1/6$, comparison with (11) yields

$$\frac{1}{3} \leq \varepsilon + \frac{128m^2\varepsilon}{s}, \quad s \leq 768m^2\varepsilon.$$

If also $\varepsilon \leq 1/(9m^3)$, then $768m^2\varepsilon \leq 256\sqrt{m\varepsilon} = 8\delta$, a contradiction. Threshold (9) implies both restrictions. Hence $s \leq 256\sqrt{m\varepsilon}$, and (5) gives

$$\|K^{\circ m} - J\|_{\text{max}} \leq 258\sqrt{m\varepsilon}. \quad (12)$$

For any unit z , its nearest m th root g satisfies

$$|z - g| \leq \frac{\pi}{2m}|z^m - 1|. \quad (13)$$

Indeed choose the reduced angle $\theta \in [-\pi/m, \pi/m]$ and use $2\sin(m|\theta|/2) \geq 2m|\theta|/\pi$. Rounding K entrywise therefore gives

$$q_0 = \|K - G\|_{\text{max}} \leq 129\pi\sqrt{\varepsilon/m} < 512\sqrt{\varepsilon/m}.$$

The threshold gives $q_0 \leq 1/(8m^2)$, so every nearest root is unique and the initial row and column stay 1.

For distinct rows a, b and $1 \leq k < m$, telescoping unit powers gives

$$\begin{aligned} \left| \sum_j (G_{aj}\overline{G_{bj}})^k \right| &\leq \varepsilon + 4mkq_0 \\ &\leq \varepsilon + \frac{m-1}{2m} < 1. \end{aligned} \quad (14)$$

Let N_r count ratios equal to ζ_m^r . Their sum is $2m$, and Fourier inversion gives the integer

$$N_r - 2 = \frac{1}{m} \sum_{k=1}^{m-1} \left[\sum_j (G_{aj}\overline{G_{bj}})^k \right] \zeta_m^{-kr}.$$

By (14) its modulus is strictly less than 1, hence it is zero. This proves the full seed condition, also when m is composite. If $\varepsilon = 0$, the exact classification in Appendix A gives $K = G$ directly. Finally

$$(H_{\text{exact}})_{ij} = \frac{H_{i0}H_{0j}}{H_{00}}G_{ij}$$

is exact and has the asserted distance. The nonexistence gap is the strict contrapositive of this construction; no distance to an empty exact set is assigned. $\square$

For the normalized residual $\rho = \max_k \|H^{\circ k}(H^{\circ k})^*/(2m) - I\|_{\text{op}}$, the hypothesis is $\rho \leq 2^{-25}m^{-4}$ and the distance bound is $512\sqrt{2\rho}$. If instead every off-diagonal *row* correlation is bounded by η , the Hermitian row-sum bound gives $\varepsilon \leq (2m-1)\eta$. No equality of row and column maximum-entry residuals is assumed.

3.2 All half-orders and certified compatibility

Theorem 3.2 (All half-orders). *For every $m \geq 2$, a unit-modulus H satisfying (9) has a dephased cyclic $\text{GH}(m, 2)$ seed G and either $T = G$ or $T = \Phi_{G,Q}(\alpha)$ for a compatible rectangle Q and a unit α , such that*

$$\|K - T\|_{\max} \leq 512\sqrt{\varepsilon/m}.$$

The construction certifies both the actual seed and compatibility. The same nonexistence residual gap holds if no seed exists at m .

Proof. For $\varepsilon = 0$ use Appendix A; assume $\varepsilon > 0$. If $s \leq 8\delta$, (5) gives (12), and the root rounding and integer Fourier certificate above are parity-free. Thus $T = G$ works.

Suppose $s > 8\delta$. Use (7) and choose $\alpha^m = x$. Divide entries on $R \times S$ by α and round all entries to μ_m , obtaining G . Put $T = \Phi_{G,R \times S}(\alpha)$. By (13),

$$q_0 = \|K - T\|_{\max} \leq 17\pi\sqrt{\varepsilon/m} < 64\sqrt{\varepsilon/m}. \quad (15)$$

The initial indices are outside R, S , so both matrices are dephased.

For two rows in the same group the block phase cancels. Each nontrivial Fourier sum of the root ratios is bounded by

$$\varepsilon + 4mkq_0 \leq \varepsilon + 256k\sqrt{m\varepsilon} < 1.$$

The integer argument certifies multiplicity two for all these pairs.

Take a cross-group pair $a \in R, b \notin R$. Active-row matching and (7) give $|d_a - x| \leq \delta/4$ and $|d_b - 1| < \delta$. The actual row-ratio product $d = d_a/d_b$ thus satisfies

$$t = |d - 1| \geq s - 5\delta/4 > s/2 \geq \delta.$$

Outside S , the ratio's m th power is within $9\delta/8$ of 1. Lemma 2.2 matches it with power error at most $\delta/16$; it cannot belong to the d -polygon because $t - \delta/16 > 9\delta/8$. The m outside entries therefore match bijectively to μ_m and the m inside entries to the other polygon. For $1 \leq k < m$, each actual restricted sum has modulus at most $128mk\varepsilon/s$.

Dividing the rectangle by α rotates one restricted sum, and rounding changes either by at most $2mkq_0$. Hence either restricted root sum in G has modulus at most

$$\frac{128mk\varepsilon}{s} + 2mkq_0 < 129k\sqrt{m\varepsilon} \leq \frac{129}{4096} < 1. \quad (16)$$

For either set of m columns, the zeroth Fourier coefficient of its root counts is m . Formula (16) makes each integer $N_r - 1$ have modulus less than 1. Thus every root occurs once on each half. This proves both that G is a seed and that the rectangle is compatible. The exact compatible-circle construction proves $T \in \mathcal{X}_m$, and (15) is stronger than required. Undoing dephasing preserves distance, and the gap again follows by contraposition. $\square$

3.3 Linear stability away from root intersections

Proposition 3.3. *Under (9), if $s > 256\sqrt{m\varepsilon}$, the circle point constructed above satisfies*

$$\|K - T\|_{\max} \leq 512\varepsilon/s. \quad (17)$$

Proof. Put $h_0 = 64m\varepsilon/s$. In the coarse rectangle, active row products are within $\delta/4$ of x and active column products within $\delta/8$ of x , so their separations from 1 are at least $s/2$. Reapply Lemma 2.2: active rows and columns now have phase-power matching error at most $2h_0$, while the original maximizing row has error at most h_0 . The separated coarse supports cannot switch.

For $j \in S$, comparison at the maximizing row gives $|c_j - x| \leq 3h_0$, hence $|L_{ij} - x| \leq 5h_0$ on the rectangle. Both off-rectangle strips have $|L_{ij} - 1| \leq 2h_0$. For a top-left entry (b, j) , compare row b with the maximizing row a . The ratio product x/d_b has separation at least $s - \delta > s/2$. Its m th power at j is within $9\delta/8$ of 1 by the coarse estimate, and therefore belongs to the one-polygon. Improved matching bounds its error by $2h_0$. Since $|L_{aj} - 1| \leq h_0$, we get $|L_{bj} - 1| \leq 3h_0$.

The whole power-pattern error is at most $5h_0 = 320m\varepsilon/s$. Equation (13) yields

$$\|K - T\|_{\max} \leq \frac{\pi}{2m} 5h_0 = 160\pi\varepsilon/s < 512\varepsilon/s.$$

Nearest-root uniqueness at the coarse threshold identifies the same seed and rectangle as in Theorem 3.2. $\square$

At a nonroot exact point $s = |\alpha^m - 1| > 0$. Thus the deterioration of the explicit local linear constant is displayed as one approaches a root intersection.

3.4 A genuine square-root obstruction at order four

Proposition 3.4. *Let*

$$G = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad K(t) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -e^{it} & e^{2it} & -e^{it} \\ 1 & e^{2it} & -e^{it} & -e^{it} \\ 1 & -e^{it} & -e^{it} & e^{2it} \end{pmatrix}.$$

Then

$$\varepsilon(K(t)) = 6(1 - \cos t) \leq 3t^2, \quad \text{dist}_{\max}(K(t), \mathcal{X}_2) \geq 2t/\pi \quad (0 < t \leq 1/4).$$

Consequently no exponent larger than 1/2 can replace the square root uniformly over the class containing even half-orders, already at $m = 2$.

Proof. Before dephasing, multiply entries $(0, 0), (1, 2), (2, 1), (3, 3)$ of G by e^{it} and swap columns 1, 2. The resulting matrix is $A + (e^{it} - 1)I$, where A is real symmetric Hadamard with diagonal all one and eigenvalues $2, 2, 2, -2$. Its Gram error is $2(\cos t - 1)(A - I)$, of norm $6(1 - \cos t)$. Undoing the permutation and dephasing gives the displayed $K(t)$ and preserves the residual.

Every exact circle through G changes only one 2×2 rectangle. At least one of the nine noninitial cells is outside it, and every such cell has phase t or $2t$ in $K(t)$. Its distance is at least $2t/\pi$.

A circle not through G has distance at least $\sqrt{2}$ from G . Outside its rectangle the seed signs are fixed, and a disagreement costs 2. If there is no such disagreement, either all seed-to- G ratios on the rectangle are the same sign, so the circle contains G , or both signs occur. In the latter case $\max(|\alpha - 1|, |\alpha + 1|) \geq \sqrt{2}$ for every unit α . Since $\|K(t) - G\|_{\max} \leq 2t$, a nonincident circle has distance at least $\sqrt{2} - 2t > 2t/\pi$ in the stated range. Other root points are separated as well. Exhaustiveness of the finite-circle classification now proves the distance lower bound. $\square$

This does not prove square-root sharpness at odd half-order. The order-four family and its intersecting branches are classical; no historical priority is asserted for this residual curve.

Scalar sharpness is insufficient for matrix sharpness. There is also a scalar square-root example: take two copies of each m th root, keep both copies of 1 fixed, and replace the two copies of another root ζ_m by $\zeta_m e^{it}, \zeta_m e^{-it}$. The moment errors are $2\zeta_m^k (\cos(kt) - 1) = O_m(t^2)$. The bottleneck distance to a union of two regular m -gons is bounded below by a positive multiple of $|t|$ for small t : unchanged roots force both rotation parameters close to zero, whereas the split root forces one a distance comparable to t away. This scalar obstruction alone says nothing about simultaneous matrix row and column conditions; those may remove the first-order split directions.

4 An integer tangent matrix and explicit inverse bounds

Fix odd $m \geq 3$, $n = 2m$, and a labelled dephased seed $G_{aj} = \zeta_m^{E_{aj}}$. Use real dephased phase coordinates $X_{0j} = X_{i0} = 0$; their dimension is $d = (2m - 1)^2$. For $a < b$ and $r \in \mathbb{Z}/m\mathbb{Z}$, define

$$S_{ab,r} = \{j : E_{aj} - E_{bj} = r \pmod{m}\}, \quad T_{ab,r}(X) = \sum_{j \in S_{ab,r}} (X_{aj} - X_{bj}).$$

Each class has two elements. Let the integer map L_G have coordinates

$$(L_G X)_{ab,r} = T_{ab,r}(X) - T_{ab,0}(X), \quad 1 \leq r < m.$$

It has $n(n-1)(m-1)/2$ rows and d columns. Every coefficient is 0, 1, or -1 , with at most eight nonzero coefficients in a row.

Lemma 4.1. *The kernel of L_G is exactly the dephased simultaneous first-order kernel of the power-Hadamard equations.*

Proof. For $K(t) = G \circ e^{itX}$, the derivative of the (a, b) Gram entry at power k is

$$ik \sum_r \zeta_m^{kr} T_{ab,r}(X).$$

All these derivatives vanish for $1 \leq k < m$ exactly when the nontrivial Fourier coefficients of $(T_{ab,r})_r$ vanish, equivalently that vector is constant. This is $L_G X = 0$. Diagonal Gram entries are identically n , and row equations suffice. $\square$

This proof works for composite m as well. The kernel is an intersection of power tangent kernels; it is not automatically the ordinary first-power defect for $m > 3$.

Proposition 4.2 (Inverse-minor conditioning). *Suppose $\text{rank } L_G = d$. Choose any invertible $d \times d$ row submatrix B , and put $C = \|B^{-1}\|_{\infty \rightarrow \infty}$. For a dephased unit-modulus matrix $K = G \circ e^{iX}$ with*

$$x = \|X\|_{\max} \leq \frac{1}{8Cm(m-1)},$$

one has

$$\|K - G\|_{\max} \leq x \leq \frac{4Ch}{m} \varepsilon. \quad (18)$$

Moreover $C \leq d8^{(d-1)/2}$. If G is obtained by Theorem 3.1, the phase-radius condition follows upon also imposing

$$\varepsilon \leq \frac{1}{2^{26} C^2 m(m-1)^2}. \quad (19)$$

Proof. For a row pair put $Y_j = X_{aj} - X_{bj}$, so $|Y_j| \leq 2x$. Since exact seed sums vanish and $|e^{iu} - 1 - iu| \leq u^2/2$,

$$\left| \sum_j \zeta_m^{k(E_{aj} - E_{bj})} Y_j \right| \leq \frac{\varepsilon}{k} + 2nkx^2.$$

Fourier inversion and $|\zeta_m^{-kr} - 1| \leq 2$ give

$$\|L_G X\|_{\max} \leq \frac{2h}{m}\varepsilon + 2n(m-1)x^2 = \frac{2h}{m}\varepsilon + 4m(m-1)x^2.$$

Since $x \leq C \|L_G X\|_{\max}$, the phase-radius condition absorbs the last term into $x/2$ and proves (18).

The integer determinant of B has modulus at least 1. Every cofactor has modulus at most $8^{(d-1)/2}$ by Hadamard's determinant inequality, so summing a row of the inverse proves the stated bound on C . Finally Theorem 3.1 and the principal-angle estimate give

$$x \leq \frac{\pi}{2} \|K - G\|_{\max} \leq 1024 \sqrt{\varepsilon/m}.$$

Condition (19) makes this at most $1/[8Cm(m-1)]$. $\square$

The seed-independent inverse-minor bound is deliberately crude and is not polynomial in m ; the graph argument below avoids this cost when the graph is connected.

Proposition 4.3 (Odd-seed local dichotomy). *At an odd-half-order seed, the best local residual exponent is 1 if L_G is injective and $1/2$ otherwise, in the precise local sense of Definition 9.1 below. This does not assert existence of a rank-deficient odd seed.*

Proof. If $0 \neq X \in \ker L_G$, normalize $\|X\|_{\max} = 1$ and set $K(t) = G \circ e^{itX}$. The linearized moments vanish. The preceding Taylor bound, followed by $\|M\|_{\text{op}} \leq n \|M\|_{\max}$, gives

$$\varepsilon(K(t)) \leq 2n^2(m-1)^2 t^2 = 8m^2(m-1)^2 t^2. \quad (20)$$

For sufficiently small t , G is the unique nearest exact dephased solution: at odd m the exact set consists of root matrices, and any different root matrix is separated from G by at least $2 \sin(\pi/m)$. Meanwhile

$$\|K(t) - G\|_{\max} = 2 \sin(|t|/2) \geq 2|t|/\pi \quad (|t| \leq \pi).$$

No exponent greater than $1/2$ is possible, and Theorem 3.1 supplies $1/2$. If L_G is injective, Proposition 4.2 supplies exponent 1. Perturbing one noninitial entry by e^{it} gives distance of order $|t|$ from the finite root set and residual at most a constant times $|t|$, excluding exponents greater than 1. $\square$

4.1 Exact finite certificates

The supplied script checks every GH count, constructs L_G , selects a full-rank row minor by exact rational elimination, and verifies $BB^{-1} = I$. No floating-point rank threshold is used. For the classical cubic order-six seed it gives shape 30×25 , rank 25, determinant 34992, and $C = 18$. Thus $x \leq 1/864$ yields $\|K - G\|_{\max} \leq 36\varepsilon$.

For the order-ten BH(10, 5) seed in McNulty–Weigert's Appendix A [4], it gives shape 180×81 , rank 81, determinant 19531250, and $C = 503/5$. Thus $x \leq 1/16096$ yields $\|K - G\|_{\max} \leq (503/3)\varepsilon$. Appendix B records the matrices and certificate convention; the exact pivot lists are in the companion JSON. These certificates supply the particular inverse-minor constants, not a discovery of isolation of the classical seeds, whose zero ordinary defects were already known.

5 The rectangle graph and polynomial conditioning

Here $m \geq 2$ need not be odd. For a cyclic $\text{GH}(m, 2)$ seed, put a vertex at each cell (a, j) and join (a, j) to (b, ℓ) when

$$a \neq b, \quad j \neq \ell, \quad G_{aj}G_{b\ell} = G_{a\ell}G_{bj}. \quad (21)$$

Each row-ratio class has two columns, so the other diagonal $(a, \ell), (b, j)$ is also joined. The column GH counts follow from all-power column orthogonality and Fourier inversion. Thus every vertex has exactly one neighbor in each other row and exactly one in each other column: the graph is simple and $(n - 1)$ -regular.

Lemma 5.1. *Every component C contains the same number c_C of cells in each row and each column. It has nc_C vertices, and $\sum_C c_C = n$.*

Proof. The neighbor correspondence between any two rows is a bijection of their cells. It restricts to a bijection on cells belonging to any fixed component, since edges cannot leave it. The column argument is identical, and total-size counting identifies the row and column constants. $\square$

5.1 The component formula for the full kernel

Write V_G for the simultaneous dephased first-order kernel. Real phase directions are taken modulo row and column phase directions. Each class has a unique double-centered representative X , with every row and column sum zero, obtained by subtracting row and column means and adding the global mean. This preserves the tangent equations because phase gauges are exact symmetries.

Theorem 5.2. *A double-centered real matrix is in the simultaneous first-order kernel exactly when it is constant on every rectangle-graph component. Consequently, if the graph has r components,*

$$\dim V_G = r - 1. \quad (22)$$

In particular L_G has full column rank exactly when the graph is connected.

Proof. Finite Fourier inversion of the first-order row equations says that the sums of $X_{aj} - X_{bj}$ over the two-element ratio classes are equal. Their total is zero by double centering, so each is zero. The corresponding column assertion holds as well: differentiating square unitarity shows that a vanishing row Gram derivative at an exact square Hadamard matrix implies a vanishing column Gram derivative.

For a rectangle in (21), the row and column equations are

$$X_{aj} + X_{a\ell} - X_{bj} - X_{b\ell} = 0, \quad X_{aj} + X_{bj} - X_{a\ell} - X_{b\ell} = 0.$$

Adding and subtracting gives $X_{aj} = X_{b\ell}$ and $X_{a\ell} = X_{bj}$. Thus the values are component-constant. Conversely diagonal constancy gives these equations, hence all row derivatives by Fourier inversion.

If the component values are u_C , Lemma 5.1 makes every row and column sum equal to $\sum_C c_C u_C$. Double centering imposes precisely this one independent equation, proving (22). Dephased and double-centered spaces are complements to the same $(2n - 1)$ -dimensional gauge space. $\square$

For a proper component indicator W , the matrix $W - (c_C/n)J$ is double-centered and its dephased representative is

$$X_{ij} = W_{ij} - W_{i0} - W_{0j} + W_{00}.$$

Thus a disconnected graph supplies rational kernel directions without solving a cyclotomic linear system.

5.2 Edge estimates and graph diameter

Suppose the graph is connected, of diameter D . Write $K = G \circ e^{iX}$ in small dephased principal phases, let Y be its double centering, and put $y = \|Y\|_{\max}$. The corresponding phase-gauged $K' = G \circ e^{iY}$ has the same residual.

Lemma 5.3. *For every edge $(a, j) \sim (b, \ell)$,*

$$|Y_{aj} - Y_{b\ell}| \leq \frac{h}{m}\varepsilon + n(m-1)y^2. \quad (23)$$

Proof. Taylor expansion of the power- k row correlation gives

$$\left| \sum_t (G_{at}/G_{bt})^k (Y_{at} - Y_{bt}) \right| \leq \varepsilon/k + 2nky^2.$$

The zero-frequency sum vanishes by double centering. Fourier inversion for the two-element class j, ℓ yields

$$|Y_{aj} + Y_{a\ell} - Y_{bj} - Y_{b\ell}| \leq (h/m)\varepsilon + n(m-1)y^2.$$

By (2), the column-pair sum has the same bound. Add the two sums and divide by two to obtain (23). $\square$

Since Y has mean zero, its maximum absolute value is at most its range. Graph paths therefore give

$$y \leq D[(h/m)\varepsilon + n(m-1)y^2].$$

If $y \leq 1/[2Dn(m-1)]$, absorb the quadratic term to get $y \leq 2Dh\varepsilon/m$. Recovering dephased phases by $X_{ij} = Y_{ij} - Y_{i0} - Y_{0j} + Y_{00}$ proves

$$\|K - G\|_{\max} \leq \|X\|_{\max} \leq \frac{8Dh}{m}\varepsilon. \quad (24)$$

For completeness, a connected graph on n^2 vertices of minimum degree $n-1$ has diameter at most $3n-1$. On a geodesic of length D , select positions $0, 3, 6, \dots, 3\lfloor D/3 \rfloor$. Their closed neighborhoods are disjoint, since overlap would connect two selected vertices within distance two. Each neighborhood has at least n vertices, so

$$[\lfloor D/3 \rfloor + 1]n \leq n^2, \quad D \leq 3n-1 < 6m.$$

Applied inside a component of row weight c_C , the same argument gives component diameter at most $3c_C - 1$.

5.3 An explicit residual-only entrance condition

Now take odd $m \geq 3$ and round by Theorem 3.1. The principal phases satisfy

$$\|X\|_{\max} \leq 1024\sqrt{\varepsilon/m}, \quad y \leq 4\|X\|_{\max} \leq 4096\sqrt{\varepsilon/m}.$$

For the rounded seed's graph, the local radius follows from

$$\varepsilon \leq \frac{1}{2^{28}D^2m(m-1)^2}. \quad (25)$$

Since $D < 6m$, the simpler sufficient restriction is $\varepsilon \leq 2^{-34}m^{-5}$, already below the global threshold.

Theorem 5.4 (Polynomial linear certificate). *For odd $m \geq 3$, let H have exactly unit-modulus entries and residual at most $2^{-34}m^{-5}$. If the rectangle graph of its rounded dephased seed G is connected, then*

$$\|K - G\|_{\max} \leq 48H_{m-1}\varepsilon.$$

The graph is computable by integer modular comparisons and traversal. The theorem is a per-seed certificate; it is not a proof that every odd seed's graph is connected. A disconnected odd seed would instead yield a genuine square-root obstruction by Proposition 4.3. Discreteness of the exact set alone does not imply a zero tangent kernel.

For the two classical seeds of Section 4, exact traversal gives one component and diameters 3 and 4. Formula (24) improves their local constants to 12 and $40/3$, respectively. These are conditioning improvements, not new isolation results.

6 Component-weight obstructions at odd half-order

Let W be the zero-one indicator of any union of rectangle-graph components of a seed G . Its row and column weights equal the same integer c .

Lemma 6.1. *Any two distinct row supports of W have even intersection; the same holds for columns. In particular, over $\mathbb{F}_2$,*

$$WW^T = (c \bmod 2)I.$$

Proof. For distinct rows a, b , the ratio pairing sends a common-support column j to its unique other equal-ratio column ℓ . Diagonal component invariance gives $W_{a\ell} = W_{bj} = 1$ and $W_{b\ell} = W_{aj} = 1$, partitioning the common support into pairs. The column proof is identical; the diagonal of the displayed product is $c \bmod 2$. $\square$

Proposition 6.2. *A union of components with row weight $c = 1$ or $c = 2$ forces $n = 2m$ to be a square. Thus these weights are impossible when m is odd.*

Proof. Put $A = G \circ W$. Pairing the two columns of every ratio class and swapping them shows that the corresponding sums weighted by W_{aj} and W_{bj} coincide. Hence $AG^* = GA^*$, so

$$P = AG^*/\sqrt{n}$$

is Hermitian. From $GG^* = G^*G = nI$ and $AG^* = GA^*$ one obtains $G^*A = A^*G$, and consequently

$$P^2 = AA^*, \quad \text{tr } P = c\sqrt{n}. \tag{26}$$

If $c = 1$, then A is a unitary monomial matrix. Thus $P^2 = I$, its eigenvalues are ± 1 , and (26) makes $\sqrt{n}$ an integer.

If $c = 2$, even intersections force overlapping row supports to coincide. Column weight two then makes W , after row and column permutations, a block-diagonal union of 2×2 all-ones blocks. The corresponding G block has rank one: the two common columns must be paired with each other by the equal-ratio pairing. Each block of A has singular values $2, 0$. Thus the spectrum of P lies in $\{0, 2, -2\}$, and $\text{tr } P = 2\sqrt{n}$ again forces $\sqrt{n}$ to be an integer. Odd m gives $n \equiv 2 \pmod{4}$, which is not a square. $\square$

The Hermitian structure is produced by the component relation and the factor A ; no assertion that an arbitrary complex Hadamard matrix can be made Hermitian is used.

Proposition 6.3. *For odd m , no union of components has row weight $c \equiv 3 \pmod{4}$.*

Proof. For odd c , Lemma 6.1 makes W invertible over $\mathbb{F}_2$. Write its rows as w_i . Inclusion-exclusion for XOR Hamming weight gives, for $x \in \mathbb{F}_2^n$,

$$\text{wt} \left(\sum_i x_i w_i \pmod{2} \right) \equiv c \text{wt}(x) \pmod{4}.$$

Indeed pair intersections are even, annihilating their coefficient-two terms modulo four, and all higher terms have coefficients divisible by four. Since $x \mapsto xW$ is a bijection,

$$(1 + i)^n = \sum_x i^{\text{wt}(xW)} = (1 + i^c)^n.$$

If $c \equiv 3 \pmod{4}$, this gives $(1 + i)^n = (1 - i)^n$, hence $i^n = 1$ and $4 \mid n$, contradicting odd m . $\square$

The underlying binary quadratic-form/Gauss-sum identity is elementary and is not claimed as novel.

Corollary 6.4. *At odd half-order, every component has row weight at least four. Every cyclic GH(3, 2) has connected rectangle graph and zero simultaneous tangent kernel, without a seed classification.*

Proof. Weights 1, 2 are excluded by Proposition 6.2, and weight 3 by Proposition 6.3. A disconnected graph at $m = 3$ would have a component of weight at most three because the weights sum to six. $\square$

The exclusion is exactly $c \in \{1, 2\}$ or $c \equiv 3 \pmod{4}$; it does *not* exclude the entire residue classes one and two. For instance the abstract partitions $4 + 6$ and $5 + 5$ at $m = 5$ remain possible under these restrictions.

Because every union obeys Proposition 6.3, no component of weight one modulo four can coexist with one of weight two modulo four, and there are at most two components of weight one modulo four. Since the total weight is two modulo four, either exactly two weights are one modulo four and all others are zero modulo four, or all weights are even with an odd number equal to two modulo four. These are necessary conditions, not a construction or classification. The weight-one spectral obstruction was independently suggested during the original collaborative development; the full argument, weight-two extension, and binary obstruction are preserved here.

7 The classical quadratic prime-order family

This section proves a uniform graph and conditioning theorem for known MUB seeds. It does not construct a new seed family or assert equivalence of the entire parameter family with Butson's original construction.

7.1 Exponent formula and exact counts

Let p be an odd prime, let χ be its quadratic character, and choose

$$r \in \mathbb{F}_p \setminus \{0, 1\}, \quad \chi(1 - r) = -1, \quad q = 1 - r.$$

Rows are indexed by top labels $x \in \mathbb{F}_p$ and bottom labels $y \in \mathbb{F}_p$; columns by left labels $u \in \mathbb{F}_p$ and right labels $v \in \mathbb{F}_p$. All fractions below are in $\mathbb{F}_p$. Define

$$\begin{aligned} E_{Tx, Lu} &= xu, & E_{Tx, Rv} &= rx^2 + xv, \\ E_{By, Lu} &= (u^2 - 2uy)/4, & E_{By, Rv} &= (v - y)^2/(4q) - y^2/4. \end{aligned} \tag{27}$$

Then $G = \zeta_p^E$ is dephased at top row $x = 0$ and left column $u = 0$.

Two distinct top rows have nonzero linear difference on each block, hence each residue occurs once per block. The same holds for distinct bottom rows. For a top row x and bottom row y , the differences are

$$C - (u - u_0)^2/4, \quad C - (v - v_0)^2/(4q),$$

where $u_0 = 2x + y$, $v_0 = 2qx + y$, and $C = (2x + y)^2/4$. Since $\chi(q) = -1$, the noncentral quadratic value sets are complementary, with multiplicity two on the block where they occur; the common central value occurs once in each block. Thus every residue occurs twice, proving the cyclic $\text{GH}(p, 2)$ condition.

7.2 Explicit attribution bridge from MUB blocks

Let $F_{xu} = p^{-1/2}\zeta_p^{xu}$, $D_a = \text{diag}(\zeta_p^{ax^2})$, and $U_a = D_a F$. The standard MUB block construction [4, Theorem 3 and its two-block specializations] gives the normalized matrix

$$\frac{1}{\sqrt{2}} \begin{pmatrix} F & U_r \\ U_1^* F & -U_1^* U_r \end{pmatrix}.$$

For $a \neq 0$ put $\gamma(a) = p^{-1/2} \sum_x \zeta_p^{ax^2}$. Completing the square and using $\gamma(a) = \chi(a)\gamma(1)$ shows that, after multiplying the displayed matrix by $\sqrt{2p}$, its lower blocks have entries

$$\gamma(-1)\zeta_p^{(u-y)^2/4}, \quad -\gamma(-q)\zeta_p^{(v-y)^2/(4q)}.$$

The first top row is already one. The first-column entry in bottom row y is $\gamma(-1)\zeta_p^{y^2/4}$. Dividing that row by this entry produces the lower exponents in (27); the residual right-block constant is

$$-\gamma(-q)/\gamma(-1) = -\chi(q) = 1.$$

The top exponents are unchanged. This gives all four claimed blocks, explicitly identifying the construction and dephasing. The Gauss-sum formulation is also discussed in [5, Section 4]. Here “Butson-type” only specifies the root level, not an unproved whole-family equivalence with Butson’s original construction.

7.3 Within-group components

The graph induced on top-row cells splits into the p sets

$$T_s = \{(Tx, Lu) : u - rx = s\} \cup \{(Tx, Rv) : v + rx = s\}.$$

Each is $K_{p,p}$ minus a perfect matching, with the omitted pair in the same row. Indeed, for $x \neq x'$, equal-ratio columns satisfy $v = u - r(x + x')$. Since $p \geq 3$, each T_s is connected with diameter at most three.

Similarly the bottom induced graph splits into

$$B_t = \{(By, Lu) : qu + (r/2)y = t\} \cup \{(By, Rv) : v - (r/2)y = t\}.$$

For $y \neq y'$, paired columns satisfy $v = qu + (r/2)(y + y')$, again giving $K_{p,p}$ minus a perfect matching and diameter at most three. The conditions $r \neq 0$, $q \neq 0$ ensure the required distinct vertices in each part.

7.4 Cross-group neighbors and diameter seven

For a fixed top-left cell (Tx, Lu) , its neighbor in bottom row y lies in the component with label

$$t = 4qx - qu + \frac{4 - 3r}{2}y. \quad (28)$$

For a fixed top-right cell (Tx, Rv) , that label is

$$t = 4qx - v + \frac{4 - r}{2}y. \quad (29)$$

To verify the formulas away from the central value, use the reflections $u' = 4x + 2y - u$ and $v' = 4qx + 2y - v$ in the completed-square differences. At the central root the pairing switches blocks, but both centers have the same component label, so the formulas remain valid.

The two slopes in (28)–(29) differ by $r \neq 0$, so at least one column type can reach every bottom component in one cross-group step. A vertex of the other type reaches a good type in one within- T_s edge. Thus every top vertex is within two steps of every prescribed bottom component.

Conversely, for a fixed bottom-left or bottom-right cell the labels of its top neighbors are, respectively,

$$s = (4 - r)x + 2y - u, \quad s = (4 - 3r)x + 2y - v. \quad (30)$$

Their slopes differ by $2r \neq 0$. Therefore every bottom vertex lies within two steps of each prescribed top component. Two top vertices can reach the same bottom component in at most two steps each and connect inside it in three; the same applies to bottom vertices. Opposite-group vertices need at most five steps.

Theorem 7.1. *For every odd prime and every admissible parameter in (27), the rectangle graph is connected and has diameter at most seven.*

The argument uses elementary finite-field counts and is uniform; it needs neither numerical rank computations nor a classification of seeds.

Corollary 7.2. *Suppose the rounded seed of a unit-modulus matrix of order $2p$ is a matrix in (27), or is equivalent to one by row/column phases and permutations. If*

$$\varepsilon \leq 2^{-34}p^{-3},$$

then

$$\|K - G\|_{\max} \leq \frac{56H_{p-1}}{p}\varepsilon.$$

Every seed in this family has zero simultaneous dephased tangent kernel and best local exponent one.

Proof. The rectangle condition and diameter are invariant under the stated equivalences. The global entrance bound applies at the given threshold, and (25) follows since $D \leq 7$ and $49 < 64$. Formula (24) gives the result. The kernel and exponent assertions follow from Theorem 5.2 and Proposition 4.3. $\square$

Initial lightweight checks verified counts and connectivity for all admissible parameters at $p = 3, 5, 7, 11, 13, 17, 19$. The independent analytical review additionally checked all 45 admissible parameters through $p = 23$, including the adjacency formulas; for $p \leq 7$ it traversed all source vertices for exact diameters. None of these finite checks is used in the general proof.

Conjecture 4.5 of [5] concerns ordinary first-power isolation for specified MUB parameters. Zero ordinary defect would imply zero simultaneous defect, but ordinary isolation and zero ordinary defect differ, and our all-power theorem establishes neither ordinary assertion in general. In particular, arbitrary real tangent coefficients cannot be held fixed while a cyclotomic Galois automorphism is applied to the seed equations. Any such ordinary-defect conclusion would require additional work.

8 Second-order liftability and the exact tangent cone

Fix $m \geq 2$ and a dephased cyclic seed G . Let V_G denote its simultaneous dephased first-order kernel. Consider a twice-differentiable unit-modulus phase curve

$$K(t) = G \circ \exp(i[tX + t^2Y + o(t^2)]),$$

with real X, Y zero on the initial row and column. The coefficient Y is half the usual second phase derivative. Its phase two-jet is *admissible* if every required power Gram residual is $o(t^2)$.

Theorem 8.1. *The admissible phase two-jets are exactly those satisfying $Y \in V_G$ and either $X = 0$ or*

$$X = \beta \mathbf{1}_{R \times S}, \quad \beta \in \mathbb{R} \setminus \{0\},$$

for a compatible rectangle with $|R| = |S| = m$ avoiding the initial indices. Thus the nonzero velocities admitting any second-order lift are exactly the tangent lines of the compatible phase circles. This does not assert that every admissible two-jet extends to an exact curve with its prescribed acceleration Y .

8.1 Conjugate powers separate acceleration and drift

For a row pair put

$$g_j = G_{aj} \overline{G_{bj}}, \quad u_j = X_{aj} - X_{bj}, \quad v_j = Y_{aj} - Y_{bj}, \quad S_k = \sum_j g_j^k v_j, \quad Q_k = \sum_j g_j^k u_j^2.$$

The first-order equation is $\sum_j g_j^k u_j = 0$, and the second-order equation is

$$ikS_k - (k^2/2)Q_k = 0, \quad S_k = -(ik/2)Q_k. \quad (31)$$

Because $g_j \in \mu_m$ and u_j, v_j are real,

$$S_{m-k} = \overline{S_k}, \quad Q_{m-k} = \overline{Q_k}.$$

Comparing (31) at $m - k$ with its conjugate at k gives

$$-\frac{i(m-k)}{2} \overline{Q_k} = \frac{ik}{2} \overline{Q_k}.$$

Thus $Q_k = S_k = 0$ for all nontrivial characters. When $k = m - k$, including $m = 2$, the same conclusion follows by separating real and imaginary parts. Hence acceleration cannot cancel quadratic phase drift: admissibility forces separately

$$\sum_j g_j^k u_j^2 = 0, \quad \sum_j g_j^k v_j = 0. \quad (32)$$

The latter is $Y \in V_G$. The same reasoning applies to columns because row and column Gram operator norms coincide at each t .

8.2 A real distance-geometry lemma

Subtract linear row and column phase gauges to make X double-centered; admissibility is preserved. For each row pair, Fourier inversion says that the first-order differences have zero sum in each two-element class, hence are $u, -u$ there. The squared sums in (32) are constant across classes, so

$$(X_{aj} - X_{bj})^2 \text{ is independent of } j, \quad (X_{aj} - X_{al})^2 \text{ is independent of } a. \quad (33)$$

Lemma 8.2. *A nonzero double-centered real matrix satisfying (33) is $X = ars^\top$, where $a \neq 0$ and r, s are sign vectors each having m entries of each sign.*

Proof. Choose a nonconstant column u , which exists by nonzero column centering. The first equality in (33) makes every other labelled column configuration on the real line have the same pairwise distances as u . Two such finite configurations differ only by a translation and a reflection: fix two distinct coordinates as anchors, then solve their two squared-distance equations for each remaining coordinate. Centering removes translation, so each column is u or $-u$. Row centering forces exactly m columns of each sign. Apply the second equality in (33) to an opposite-sign pair: every coordinate of u has the same absolute value. Column centering gives exactly m coordinates of each sign. $\square$

8.3 Dephasing and compatibility

The dephased representative of $ars^\top$ is

$$X_{ij} = a(r_i - r_0)(s_j - s_0).$$

It is zero off $R \times S$, where $R = \{i : r_i \neq r_0\}$ and $S = \{j : s_j \neq s_0\}$, and equals $\beta = 4ar_0s_0 \neq 0$ on that rectangle. Both sets have size m and exclude the initial indices. For $a \in R, b \notin R$, its first-order equation is

$$\sum_{j \in S} (G_{aj} \overline{G_{bj}})^k = 0 \quad (1 \leq k < m).$$

There are m roots in this multiset. Fourier inversion says each root occurs once, precisely compatibility; the complementary half follows from the full counts. Conversely compatibility makes $G \circ e^{it\beta \mathbf{1}Q}$ an exact curve. Its linear and quadratic coefficients vanish, while every $Y \in V_G$ contributes zero at second order. Together with (32), this proves Theorem 8.1.

8.4 A graph test for branches

Let the graph components be $C_1, \dots, C_r$ with row weights $c_1, \dots, c_r$, where $(0, 0) \in C_1$. Compatible rectangles correspond exactly to sign assignments $\sigma_\ell \in \{1, -1\}$ satisfying

$$\sigma_1 = 1, \quad \sum_{\ell} c_\ell \sigma_\ell = 0, \quad (34)$$

for which the component-constant sign matrix has real rank one. Indeed a balanced rank-one sign matrix is $rs^\top$; normalizing its $(0, 0)$ value to one removes sign ambiguity and dephasing gives the rectangle. Conversely the double-centered rectangle velocity is a scalar multiple of that matrix. The graph-kernel theorem gives both implications. This is an exact finite sign test followed by a rank-one test.

The number b of compatible circles through G obeys

$$b \leq \binom{r-1}{\lfloor (r-1)/2 \rfloor}. \quad (35)$$

To see this, the positive component subset contains C_1 and has total row weight m . The remaining subsets form an antichain because all weights are positive; strict inclusion increases total weight. Sperner's bound proves (35), and the rank-one condition can only reduce the number. The classical order-four root has $r = 4$ and three compatible branches, attaining the bound. The independent finite review also checked its second-order branch equations: in the three rectangle tangent coordinates (a, b, c) , the vanishing ideal is (ab, ac, bc) , the union of the three coordinate lines. This finite check is not used in the theorem.

At odd m , there are no compatible rectangles by the exact theory. Independently, the positive cells of a balanced rank-one sign matrix form a union of graph components of row weight m . Two distinct rows with the same row sign have identical support of odd size m , contradicting Lemma 6.1; there are at least two such rows since $m \geq 3$. Thus every nonzero simultaneous kernel direction at an odd seed is obstructed at second order. This does not prove that its first-order kernel is zero.

9 The best local exponent at every exact point

Definition 9.1. Fix $K_0 \in \mathcal{X}_m$. An exponent $\alpha > 0$ is valid locally if there is a neighborhood of K_0 in the unit-modulus dephased matrices and a finite constant C such that every K in that neighborhood satisfies

$$\text{dist}_{\max}(K, \mathcal{X}_m) \leq C\varepsilon(K)^\alpha.$$

The neighborhood and constant may depend on K_0 . The best local exponent is the supremum of these α .

Theorem 9.2. *At a root seed G , let r be its number of rectangle-graph components and b its number of compatible rectangles. The best local exponent equals 1 if $r = 1$, or if $r = 2, b = 1$; it equals $1/2$ in all other cases. At every nonroot exact point it equals 1.*

When $r = 2$, the sign test in (34) allows at most one compatible rectangle. Thus a smooth circle with its one-dimensional tangent kernel is distinguished from a smooth circle with additional obstructed first-order directions.

9.1 Explicit linear estimates at the two root types

Write $K = G \circ e^{iX}$ in dephased small principal phases, and put $x = \|X\|_{\max}$.

If $r = 1$, the restriction

$$x \leq 1/(128m^3) \tag{36}$$

is sufficient for

$$\text{dist}_{\max}(K, \mathcal{X}_m) \leq 48h\varepsilon. \tag{37}$$

Indeed $y \leq 4x$ and $D < 6m$, so (36) implies the absorption radius in Section 5; equation (24) then proves (37).

Suppose $r = 2, b = 1$ and let Q be the compatible rectangle. The two components are exactly the positive and negative cells of its balanced rank-one sign matrix. Each has row weight m and diameter at most $3m - 1$. In this case

$$x \leq 1/(8192hm^4) \tag{38}$$

implies that some exact circle point T through G satisfies

$$\|K - T\|_{\max} \leq 24h\varepsilon. \tag{39}$$

Here is the full quantitative transverse argument. Let D be the larger component diameter. Double-center X and subtract its two component means, which are opposites because component sizes agree. Dephase the component-constant part to write

$$X = \beta \mathbf{1}_Q + Z, \quad |\beta| \leq 16x, \quad z = \|Z\|_{\max} \leq 17x.$$

The centered version of Z has mean zero on each component. For the derivative $\mathcal{D}_G$ of the all-power row Gram map, with output norm the maximum operator norm over powers, the linear edge argument on each component gives

$$z \leq C_0 \|\mathcal{D}_G(Z)\|, \quad C_0 = 4Dh/m < 12h. \quad (40)$$

The column derivative has the same norm: at an exact Hadamard matrix it is a unitary conjugate of the row derivative. This accounts for the row-and-column edge estimate and the dephasing factor four.

Let $T = G \circ e^{i\beta \mathbf{1}_Q}$, which is exact, and let $\mathcal{F}$ denote the tuple of power Gram residuals. Direct differentiation and Taylor's inequality give

$$\begin{aligned} \|(\mathcal{D}_T - \mathcal{D}_G)(Z)\| &\leq B|\beta|z, \\ \|\mathcal{F}(T \circ e^{iZ}) - \mathcal{D}_T(Z)\| &\leq Bz^2, \quad B = 2n^2(m-1)^2 \leq 8m^4. \end{aligned}$$

For the first bound, use $|e^{ik\beta\Delta(\mathbf{1}_Q)} - 1| \leq k|\beta|$ and $|\Delta Z| \leq 2z$ entrywise, sum n terms, and bound an $n \times n$ operator norm by n times its maximum entry. The second bound uses the same quadratic Taylor remainder as before. Equation (40) now yields

$$z \leq C_0\varepsilon + C_0B(|\beta| + z)z.$$

Since $|\beta| + z \leq 33x$, (38) gives

$$C_0B(33x) < \frac{(12h)(8m^4)33}{8192hm^4} < \frac{1}{2}.$$

Absorb to obtain $z \leq 2C_0\varepsilon < 24h\varepsilon$, proving (39). These are local phase-radius estimates; no residual-only entrance condition selecting a prescribed even-order root neighborhood is asserted.

9.2 Sharpness in every other root case

The kernel dimension is $r - 1$. If $r = 2, b = 0$, every nonzero kernel vector lies outside the exact tangent cone. If $r \geq 3$, the kernel has dimension at least two while the cone is a finite union of compatible-rectangle lines, so again select X outside it. Along $K(t) = G \circ e^{itX}$ the residual is $O_{m,X}(t^2)$.

The exact local set is the finite union of incident smooth circles and the root point, by Appendix A. The chosen velocity has positive angle from every incident tangent line, so its distance from that local union is at least $c_X|t|$ for sufficiently small t . Nonincident compact exact components have a positive fixed distance from G . Thus no exponent greater than $1/2$ can hold. Theorem 3.2 supplies exponent $1/2$.

In the linear root cases, a direction transverse to the isolated point or the sole incident circle gives distance of order $|t|$ and residual $O(|t|)$, excluding exponents greater than one. This completes the root part of Theorem 9.2. It does not assert that any disconnected odd seed exists; if one is found, its lack of compatible rectangles automatically puts it in the square-root case.

9.3 Nonroot points

At a fixed nonroot point the separation s tends to $|\alpha^m - 1| > 0$. Continuity ensures both the global small-residual threshold and $s > 256\sqrt{m\varepsilon}$ in a sufficiently small neighborhood. Proposition 3.3 then gives a local linear estimate. The local exact set is one smooth arc. A transverse phase perturbation has distance $\Omega(|t|)$ and residual $O(|t|)$, ruling out exponent greater than one and completing the theorem.

Thus, at a supplied seed, the exponent is computable by forming the exact rectangle graph and counting components; only in the two-component case is the balanced rank-one sign test needed. No general Łojasiewicz exponent calculation is required.

10 Dependencies and open questions

The quantitative scalar-to-support-to-rounding chain is written out in Sections 2–3. Its parity-free part proves the all-half-order construction, while the half-integer trace is the only oddness input in the global odd theorem. The graph/kernel equivalence and its local conditioning are independent of global stability; the residual-only entrance estimate uses the global theorem. Component obstructions use graph pairings and elementary spectral and binary identities. The quadratic-family argument uses direct counts, the graph bound, and the global entrance estimate, not the two small rank computations. The second-order conjugate-power proof is independent of quantitative stability, and the local exponent classification combines these separate ingredients with exact finite-circle geometry.

Several questions remain unresolved.

1. Is the rectangle graph connected for every cyclic $\text{GH}(m, 2)$ with odd m ? Neither a proof nor a disconnected odd seed is supplied. A proof would yield the polynomial linear estimate at every odd seed; a counterexample would give sharp square-root behavior there.
2. Which abstract component-weight patterns are realized? The exclusions leave patterns such as $4 + 6$ and $5 + 5$ at $m = 5$; they are not a complete classification.
3. No new-order seed existence, including unresolved cyclic prime-power or non-prime-power existence questions, is asserted. At even half-orders, determining which seeds admit compatible rectangles remains a separate realization problem.
4. The ordinary first-power isolation conjecture and zero ordinary defect for the whole quadratic family remain outside these results.
5. Constants and dependence on m have not been optimized. The inverse-minor estimate is not polynomial, while the connected-graph estimates are.
6. Historical comparison is incomplete. In particular a full-text comparison with Craigen–Woodford [6] remains necessary before any novelty claim.

The terminology “almost Hadamard” in [7] refers to real matrices whose normalization is orthogonal and locally maximizes the entrywise 1-norm; that is a different problem from the present all-power residual. S-Hadamard matrices and square-defect questions [8] are related but require fewer powers than the full range when $m > 3$. Neither comparison justifies a priority claim. The ordinary defect framework remains that of [3].

A Attributed exact classification and local geometry

For self-contained use of the zero-residual endpoint and the local distance arguments, this appendix presents the relevant exact proof from the pinned source [1]. The root/single-support dichotomy, compatible-circle exhaustiveness, and finite-circle geometry are attributed to that source, including its declared OpenAI order-six antecedent [2]. The common-support product obstruction is also the elementary cyclic-even complete-mapping obstruction, as discussed there in connection with Hall–Paige [9] and [10]. No deep converse from that theory is used.

A.1 Exact polygons and coincident supports

Let $v_1, \dots, v_{2m}$ be unit and satisfy $\sum_j v_j^k = 0$ for $1 \leq k < m$. Newton’s identities make $e_1, \dots, e_{m-1}$ zero. The complement identity $e_{2m-k} = e_{2m}\overline{e_k}$ makes the corresponding upper coefficients vanish, so

$$\prod_j (z - v_j) = z^{2m} + (-1)^m e_m z^m + e_{2m} = (z^m - a)(z^m - b), \quad |a| = |b| = 1. \quad (41)$$

The moduli follow because the roots of both factors are among the unit v_j . Each factor has m simple roots; repeated factors are counted with multiplicity. Thus the entries form two rotated regular m -gons and their product is exactly ab .

For an exact dephased K , apply this to distinct row and column ratios. Let $L = K^{\circ m}$. Every noninitial row, compared to the initial row, is either all ones in L or has m ones and m copies of one $x \neq 1$. Call the latter m positions its support; they exclude column zero. Columns have the analogous alternative.

If two nonconstant rows have different supports S, U , put $t = |S \cap U|$. Since both have size m and exclude the initial column, $1 \leq t \leq m-1$. Their L ratio has values $1, x, y^{-1}, xy^{-1}$ on $(S \cup U)^c, S \setminus U, U \setminus S, S \cap U$, with multiplicities $t, m-t, m-t, t$. At most two values are allowed. As $x, y^{-1} \neq 1$, this forces $y^{-1} = x$, then $x^2 = 1$ and $x = y = -1$.

Each actual row has product -1 by (41), because its power values are $1, -1$. Their actual ratio also has those two distinct power values, so the same formula makes its product -1 . But its product is the quotient $(-1)/(-1) = 1$, a contradiction. Hence all nonconstant row supports coincide, and transposition proves the column statement. This argument does not require parity.

A.2 The exact balanced block alternative

If $L \neq J$, let R be its nonconstant rows and S their common support. A column in S is nonconstant precisely on R , so the column alternative gives $|R| = |S| = m$ and one common non-one value x . Consequently, after placing the constant groups first,

$$K = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad L = \begin{pmatrix} J_m & J_m \\ J_m & xJ_m \end{pmatrix}.$$

For a cross-group row or column ratio, the distinct power values split its two regular polygons by groups. Each restricted sum is zero, giving

$$A^*B = C^*D = AC^* = BD^* = 0,$$

while the diagonal Gram identities give $AA^* + BB^* = CC^* + DD^* = 2mI$. For each block V , it follows that $(VV^*)^2 = 2mVV^*$. Thus $VV^*/(2m)$ is a projection with

$$\text{rank } V = \text{tr}(VV^*/(2m)) = m/2,$$

and all its nonzero singular values are $\sqrt{2m}$. The integer rank forces m even. Hence odd m admits only the root pattern $K^{\circ m} = J$.

In the root pattern, let N_r be the root counts of a row ratio. All nontrivial Fourier coefficients vanish and the zeroth is $2m$, so Fourier inversion gives $N_r = 2$. Conversely these counts make all required power sums vanish. This applies to composite m without a primality assumption, and also gives the column counts.

A.3 Compatible circles and exhaustiveness

For a seed and compatible rectangle, same-group row ratios are unchanged when the rectangle is multiplied by α . In a cross-group pair, the power sum vanishes separately on S and S^c , because each half contains all roots once. Multiplication of one zero sum by $\alpha^{\pm k}$ preserves orthogonality. Hence every $\Phi_{G,Q}(\alpha)$ belongs to $\mathcal{X}_m$.

Conversely, in a nonroot block pattern choose $\alpha^m = x$ and put

$$G = \begin{pmatrix} A & B \\ C & \alpha^{-1}D \end{pmatrix}.$$

Its entries lie in μ_m . At every power k , cross-group row ratios of K split into two regular polygons, so, writing $V_k = V^{\circ k}$, $A_k C_k^* = B_k D_k^* = 0$. Replacing D_k by $\alpha^{-k} D_k$ leaves the diagonal Gram blocks unchanged and the off-diagonal blocks zero. Thus G is exact and is a seed by Fourier inversion. Each restricted cross-group ratio in G is one full polygon in μ_m , proving compatibility. This proves exhaustiveness.

There are finitely many root matrices and finitely many possible rectangles for fixed m . Thus $\mathcal{X}_m$ is a finite union of roots and compact circle images, possibly empty. At odd m there are no compatible circles, since one would produce $\alpha^m \neq 1$ and contradict the projection rank.

A.4 Intersections and local branches

A nonroot point on a circle has m th power 1 off its rectangle and $\alpha^m \neq 1$ on it; therefore it determines the rectangle uniquely. If

$$\Phi_{G,Q}(\alpha) = \Phi_{G',Q'}(\beta)$$

is nonroot, then $Q = Q'$, the seeds coincide outside Q , and $G' = (\alpha/\beta)G$ on Q , with $\alpha/\beta \in \mu_m$. Their entire circle images agree by reparametrization. After duplicate images are removed, distinct circles meet only at roots.

Every phase map is injective, as one rectangle entry recovers α . It has exactly m root points, corresponding to $\alpha^m = 1$. These roots divide it into m open arcs, each with distinct endpoints for $m \geq 2$. The finite graph obtained by identifying actual labelled root endpoints maps continuously and bijectively to $\mathcal{X}_m$; compactness and the Hausdorff property make it a homeomorphism. Parallel edges and isolated vertices are permitted.

The map has affine form

$$\Phi_{G,Q}(\alpha) = G + (\alpha - 1)(G \circ \mathbf{1}_Q),$$

so each image is a smooth circle with nonzero derivative $ie^{it}(G \circ \mathbf{1}_Q)$. A nonroot point lies on exactly one image and is a positive distance from the other finitely many compact images and root points. Locally it is therefore one smooth arc. A root incident with b circles has $2b$ local branches; each incident circle can use the root itself as seed and has tangent line $\mathbb{R}(iG \circ \mathbf{1}_Q)$. Distinct incident rectangles give distinct tangent lines. These actual branch lines need not fill the linearized kernel.

If c_m is the number of distinct circles and $b(G)$ the number of compatible rectangles at a seed, each circle has precisely m seed-rectangle representations, so

$$\sum_{G \text{ a seed}} b(G) = mc_m, \quad \deg(G) = 2b(G).$$

No quotient by matrix equivalence is taken; such a quotient would require separate stabilizer analysis.

A.5 Boundary and prime-order cautions

At $m = 2$, the dephased Fourier matrix $F_4 = (i^{ij})$ is Hadamard but has a nonreal entry, so odd rigidity cannot extend unchanged to even half-orders. Its invariant rectangular ratio i also prevents conversion to a real seed using only row and column phases. The classical real seed in Proposition 3.4 has the compatible rectangle $R = \{1, 3\}$, $S = \{2, 3\}$ (zero-based labels), and produces a classical order-four phase family. For each cross-group pair, the two ratios on S are $1, -1$, directly verifying compatibility. No general existence at even $m > 2$ follows.

For odd prime p , an exact first-power Hadamard matrix with entries in μ_p automatically has each ratio root twice: its count polynomial $f(X) = \sum_{r=0}^{p-1} N_r X^r$ vanishes at ζ_p and has degree at most $p - 1$, so it is a rational multiple of $\Phi_p(X) = 1 + \dots + X^{p-1}$. Its coefficient sum is $2p$, fixing the multiple to two. This argument concerns integer counts, not arbitrary real tangent coefficients. For composite odd m , a single Butson condition does not replace the full power range here. At $m = 3$, F_6 fails the second-power requirement because its square has repeated rows. At $m = 4$, our range also requires power three and therefore cannot establish order-eight S-Hadamard nonexistence from assumptions on powers one and two alone.

B Exact finite certificate data

The files `check_tangent_rank.py` and `tangent_rank_results.json` in the companion `certificates` directory are byte-for-byte copies of the reviewed revision. The script uses Python and SymPy exact arithmetic. Its variables are ordered lexicographically as (a, j) with $1 \leq a, j < n$, and its rows are ordered by $a < b$ and then $1 \leq r < m$. Thus the JSON pivot indices specify the chosen row minors unambiguously. The script verifies all two-element ratio classes and the identity $BB^{-1} = I$ before recording the determinant and inverse infinity norm. Running the script writes its adjacent JSON; use a temporary copy if preserving the supplied certificate bytes is important.

The exponent matrices used are the classical cubic seed from the attributed exact source and the order-ten seed of McNulty–Weigert [4, Appendix A]:

$$E_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 2 & 1 \\ 0 & 1 & 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 0 & 1 & 2 \\ 0 & 2 & 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

$$E_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 4 & 2 & 4 & 1 & 3 & 0 \\ 0 & 2 & 4 & 1 & 3 & 3 & 2 & 1 & 0 & 4 \\ 0 & 3 & 1 & 4 & 2 & 3 & 4 & 0 & 1 & 2 \\ 0 & 4 & 3 & 2 & 1 & 2 & 0 & 3 & 1 & 4 \\ 0 & 3 & 2 & 2 & 3 & 0 & 1 & 4 & 4 & 1 \\ 0 & 2 & 0 & 4 & 4 & 1 & 1 & 3 & 2 & 3 \\ 0 & 1 & 3 & 1 & 0 & 4 & 3 & 4 & 2 & 2 \\ 0 & 0 & 1 & 3 & 1 & 4 & 2 & 2 & 4 & 3 \\ 0 & 4 & 4 & 0 & 2 & 1 & 3 & 2 & 3 & 1 \end{pmatrix}.$$

The certificate records are

m	shape of L_G	rank L_G	$\det B$	$\ B^{-1}\ _{\infty \rightarrow \infty}$
3	30×25	25	34992	18
5	180×81	81	19531250	503/5

All pivot rows are retained in the JSON rather than inferred from a numerical rank. The original independent check reproduced that JSON exactly. The present manuscript preparation did not rerun mathematical searches or enlarge the finite sample. Exact graph diameters 3 and 4, and the separate quadratic-family checks reported in Section 7, are supplementary review evidence; they are not prerequisites for any general proof.

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